

Parishram 2027

Biology

DPP: 1

Principles Of Inheritance and Variation

- Q1** Which of the given genotype represent homozygous condition?
(A) Tt (B) Rr
(C) tt (D) Yy
- Q2** What is the process by which characters are passed on from parent to progeny?
(A) Inheritance (B) Variation
(C) Mendelism (D) Mutation
- Q3** The degree by which progeny differ from their parents is called:
(A) mutation. (B) inheritance.
(C) variation. (D) mendelism.
- Q4** Sahiwal cows are developed from wild cows due to-
(A) Heredity
(B) Artificial selection
(C) domestication
(D) Both (B) and (C) are correct
- Q5** Select the incorrect statement from the following
(A) Inheritance is the process by which the characters are passed from parent to the progeny.
(B) Inheritance is the basis of heredity.
(C) Variation is the degree by which the progeny differs from their parents.
(D) Sahiwal breed of buffaloes in Punjab are produced through artificial selection and domestication from wilds cows.
- Q6** Genotype is:
(A) Genetic composition of species
(B) Genetic composition of plastids
(C) Genetic composition of germ cells
(D) Genetic composition of an individual
- Q7** The ultimate biological unit which controls heredity is called:
(A) genome. (B) chromosome.
(C) genotype. (D) gene.
- Q8** An allele is:
(A) Another word for gene
(B) Homozygous genotype
(C) Heterozygous genotype
(D) One of the several possible forms of gene
- Q9** A true breeding plant is:
(A) Nearly homozygous and produces offspring of its own kind.
(B) Always homozygous recessive in its genetic constitution
(C) One that is able to breed on its own.
(D) Produced due to cross pollination among unrelated plants.
- Q10** Alleles are:-
(A) Different molecular forms of a gene.
(B) Heterozygotes
(C) Different phenotype
(D) True breeding homozygotes
- Q11** An allele is dominant if it is expressed in:
(A) Both homozygous and heterozygous states
(B) Second generation
(C) Heterozygous combination
(D) Homozygous combination



Answer Key

Q1 C
Q2 A
Q3 C
Q4 D
Q5 D
Q6 D

Q7 D
Q8 D
Q9 A
Q10 A
Q11 A



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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

A test cross is an experimental hybrid between an organism having a homozygous recessive genotype and an organism with a dominant phenotypic but unknown genotype. There are some recombinant individuals if P and Q, the paternal genes that exhibit linkage, are crossed with their respective p and q forms. Genes in incomplete linkage have a tendency to separate due to recombination because of their distant locations on the same chromosome, which increases the likelihood of them crossing over. We are finding some recombinant individuals in this query, which suggests that the paternal genes are only loosely related.

25% recombinants means 75% parental population, which is greater than 50% but less than 100%

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Q2 Text Solution:

Inheritance is the process by which **characters are passed from parent to progeny** through genetic material, ensuring the continuity of traits across generations.

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Q3 Text Solution:

A. Mutation: Mutation refers to changes in the genetic material. While mutations can cause variations, the term "mutation" specifically refers to individual genetic changes.

B. Inheritance: Inheritance refers to the passing on of genetic traits from parents to offspring, not the differences between them.

C. Variation: Variation refers to the differences between individuals of the same species, including those between progeny and their parents.

D. Mendelism: Mendelism refers to the principles of genetic inheritance proposed by Mendel, not the differences between progeny and parents.

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Q4 Text Solution:

Sahiwal cows were developed from wild cows through **artificial selection** and **domestication**.

- **Artificial selection** involves humans selectively breeding cows with desirable traits (e.g., high milk yield, disease resistance).
- **Domestication** is the long-term process where wild animals adapt to human-controlled environments.

Thus, both **artificial selection** and **domestication** played a role in developing the Sahiwal breed.

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Q5 Text Solution:

The **Sahiwal breed** is a **cattle (cow) breed**, **not a buffalo breed**. It was developed through **artificial selection and domestication** from wild **cows**, not buffaloes. The other statements (**A, B, and C**) are correct as they accurately describe **inheritance and variation**.

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Q6 Text Solution:

Genotype refers to the **genetic makeup of an individual**, including the specific alleles present in its DNA. It determines the **hereditary potential** of an organism and influences its **phenotypic traits**. For example, in pea plants, the genotype **TT (homozygous dominant)** results in a tall plant, while **tt (homozygous recessive)** results in a dwarf plant.

Video Solution:



Q7 Text Solution:

1. **Genome**: A complete set of chromosomes or genetic information in the body.

2. **Chromosome**: Small thread-like, rod-like structures which are dark in color usually appear in the nucleus are known as chromosomes. The composition of each chromosome is that it is made up of single molecule of DNA (deoxyribonucleic acid).

3. **Genotype**: This term is used for describing the actual genetic composition or genetic makeup. The phenotypes of the two plants may be similar but their genotype may be different.

4. **Gene**: This term was coined by Johannsen. A unit of genetic information that occupies a specific position on a chromosome and comes in multiple versions called alleles.

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Q8 Text Solution:

An **allele** is an **alternative form of a gene** that occurs at the **same locus** on homologous chromosomes. Different alleles can result in variations of a particular trait. For example, in pea plants, the gene for flower color has two alleles: **P (purple)** and **p (white)**. An individual can inherit **two identical alleles (PP or pp)** or **two different alleles (Pp)**.

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Q9 Text Solution:

Nearly homozygous and produces offspring of its own kind.



A true breeding plant is one that, when self-pollinated, consistently produces offspring with the same traits over multiple generations. This occurs because the plant is homozygous for the trait in question, meaning it carries identical alleles (AA or aa). True breeding is commonly used in Mendelian genetics to study inheritance patterns.

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Q10 Text Solution:

Alleles are **alternative forms of the same gene** that occur at the **same locus** on homologous chromosomes. These variations arise due to **mutations** and can lead to different traits. For example, in Mendelian genetics, the gene for **flower color in pea plants** has two alleles: **purple (dominant, P)** and **white (recessive, p)**.

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Q11 Text Solution:

An **allele is considered dominant** if it expresses its trait in **both the homozygous (AA) and heterozygous (Aa) conditions**. This means that even if only **one copy** of the dominant allele is present, the trait will be observed. For example, in **pea plants**, the **T (tall) allele** is dominant over **t (dwarf)**, so both **TT and Tt** result in a tall plant.

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